
STM32
CubeMX

1. Description

1.1. Project

Project Name	STM32CubeMX
Board Name	custom
Generated with:	STM32CubeMX 6.16.1
Date	07/12/2026

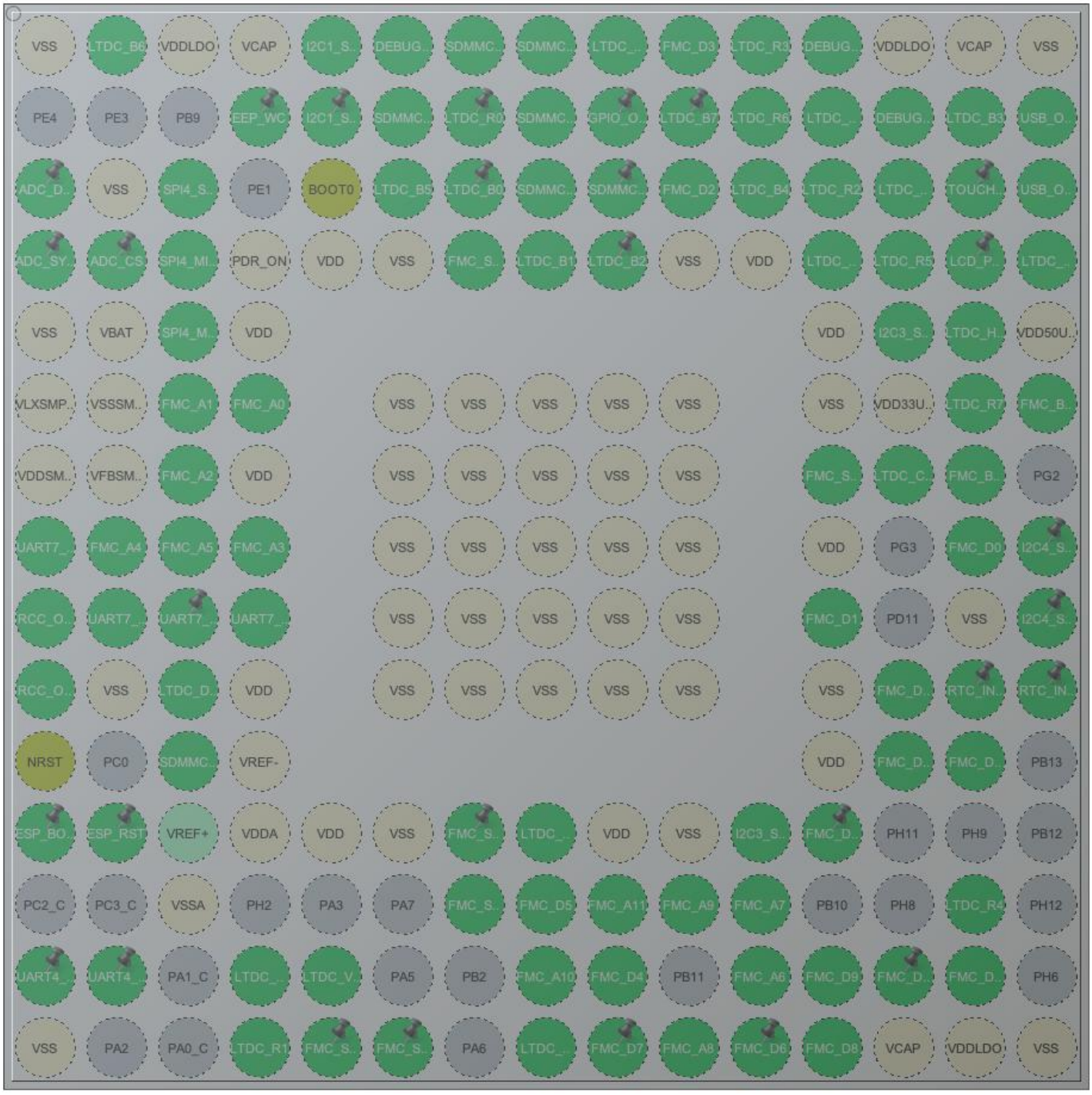
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H725/735
MCU name	STM32H725IGKx
MCU Package	UFBGA176
MCU Pin number	201

1.3. Core(s) information

Core(s)	Arm Cortex-M7

2. Pinout Configuration



UFBGA176 +25 (Top view)

3. Pins Configuration

Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSS	Power		
A2	PB8	I/O	LTDC_B6	
A3	VDDLDO	Power		
A4	VCAP	Power		
A5	PB6	I/O	I2C1_SCL	
A6	PB3(JTDO/TRACESWO)	I/O	DEBUG_JTDO-SWO	
A7	PG11	I/O	SDMMC2_D2	
A8	PG9	I/O	SDMMC2_D0	
A9	PD3	I/O	LTDC_G7	
A10	PD1	I/O	FMC_D3	
A11	PA15(JTDI)	I/O	LTDC_R3	
A12	PA14(JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
A13	VDDLDO	Power		
A14	VCAP	Power		
A15	VSS	Power		
B4	PE0 *	I/O	GPIO_Output	EEP_WC
B5	PB7	I/O	I2C1_SDA	
B6	PB4(NJTRST)	I/O	SDMMC2_D3	
B7	PG13	I/O	LTDC_R0	
B8	PD7	I/O	SDMMC2_CMD	
B9	PD5 *	I/O	GPIO_Output	
B10	PD2	I/O	LTDC_B7	
B11	PC12	I/O	LTDC_R6	
B12	PH14	I/O	LTDC_G3	
B13	PA13(JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
B14	PA8	I/O	LTDC_B3	
B15	PA12	I/O	USB_OTG_HS_DP	
C1	PC13	I/O	GPIO_EXTI13	ADC_DRDY
C2	VSS	Power		
C3	PE2	I/O	SPI4_SCK	
C5	BOOT0	Boot		
C6	PB5	I/O	LTDC_B5	
C7	PG14	I/O	LTDC_B0	
C8	PG10	I/O	SDMMC2_D1	
C9	PD4 *	I/O	GPIO_Input	SDMMC_CD
C10	PD0	I/O	FMC_D2	

Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
C11	PC11	I/O	LTDC_B4	
C12	PC10	I/O	LTDC_R2	
C13	PH13	I/O	LTDC_G2	
C14	PA10	I/O	GPIO_EXTI10	TOUCH_IRQ
C15	PA11	I/O	USB_OTG_HS_DM	
D1	PC15-OSC32_OUT *	I/O	GPIO_Output	ADC_SYNC
D2	PC14-OSC32_IN *	I/O	GPIO_Output	ADC_CS
D3	PE5	I/O	SPI4_MISO	
D4	PDR_ON	Power		
D5	VDD	Power		
D6	VSS	Power		
D7	PG15	I/O	FMC_SDNCAS	
D8	PG12	I/O	LTDC_B1	
D9	PD6	I/O	LTDC_B2	
D10	VSS	Power		
D11	VDD	Power		
D12	PH15	I/O	LTDC_G4	
D13	PA9	I/O	LTDC_R5	
D14	PC8	I/O	TIM3_CH3	LCD_PWM
D15	PC7	I/O	LTDC_G6	
E1	VSS	Power		
E2	VBAT	Power		
E3	PE6	I/O	SPI4_MOSI	
E4	VDD	Power		
E12	VDD	Power		
E13	PC9	I/O	I2C3_SDA	
E14	PC6	I/O	LTDC_HSYNC	
E15	VDD50USB	Power		
F1	VLXSMPS	Power		
F2	VSSSMPS	Power		
F3	PF1	I/O	FMC_A1	
F4	PF0	I/O	FMC_A0	
F6	VSS	Power		
F7	VSS	Power		
F8	VSS	Power		
F9	VSS	Power		
F10	VSS	Power		
F12	VSS	Power		
F13	VDD33USB	Power		

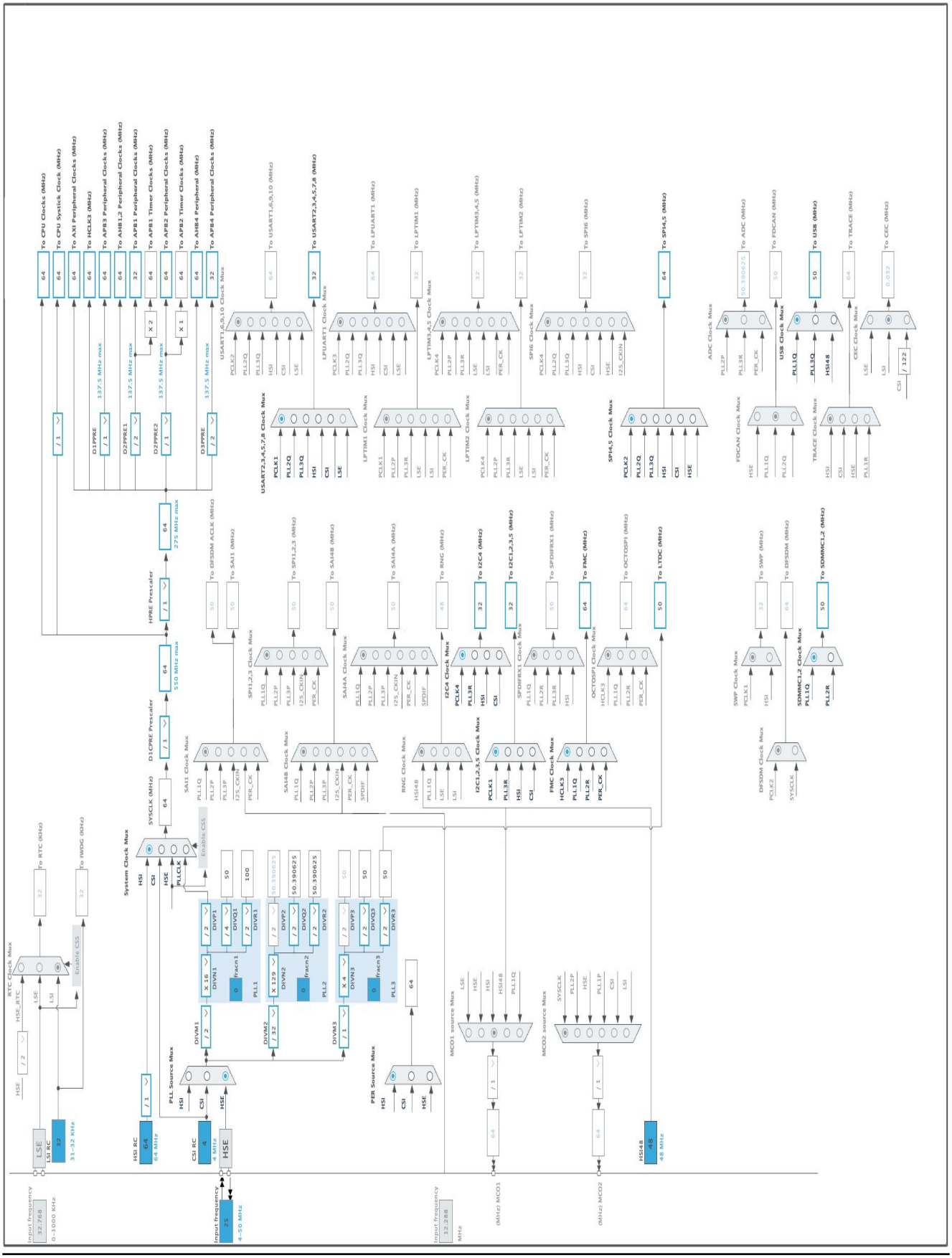
Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
F14	PG6	I/O	LTDC_R7	
F15	PG5	I/O	FMC_BA1	
G1	VDDSMPS	Power		
G2	VFBSMPS	Power		
G3	PF2	I/O	FMC_A2	
G4	VDD	Power		
G6	VSS	Power		
G7	VSS	Power		
G8	VSS	Power		
G9	VSS	Power		
G10	VSS	Power		
G12	PG8	I/O	FMC_SDCLK	
G13	PG7	I/O	LTDC_CLK	
G14	PG4	I/O	FMC_BA0	
H1	PF6	I/O	UART7_RX	
H2	PF4	I/O	FMC_A4	
H3	PF5	I/O	FMC_A5	
H4	PF3	I/O	FMC_A3	
H6	VSS	Power		
H7	VSS	Power		
H8	VSS	Power		
H9	VSS	Power		
H10	VSS	Power		
H12	VDD	Power		
H14	PD14	I/O	FMC_D0	
H15	PD13	I/O	I2C4_SDA	
J1	PH0-OSC_IN	I/O	RCC_OSC_IN	
J2	PF8	I/O	UART7_RTS	
J3	PF7	I/O	UART7_TX	
J4	PF9	I/O	UART7_CTS	
J6	VSS	Power		
J7	VSS	Power		
J8	VSS	Power		
J9	VSS	Power		
J10	VSS	Power		
J12	PD15	I/O	FMC_D1	
J14	VSS	Power		
J15	PD12	I/O	I2C4_SCL	
K1	PH1-OSC_OUT	I/O	RCC_OSC_OUT	

Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
K2	VSS	Power		
K3	PF10	I/O	LTDC_DE	
K4	VDD	Power		
K6	VSS	Power		
K7	VSS	Power		
K8	VSS	Power		
K9	VSS	Power		
K10	VSS	Power		
K12	VSS	Power		
K13	PD9	I/O	FMC_D14	
K14	PB15	I/O	GPIO_EXTI15	RTC_INTB
K15	PB14	I/O	GPIO_EXTI14	RTC_INTA
L1	NRST	Reset		
L3	PC1	I/O	SDMMC2_CK	
L4	VREF-	Power		
L12	VDD	Power		
L13	PD10	I/O	FMC_D15	
L14	PD8	I/O	FMC_D13	
M1	PC2 *	I/O	GPIO_Output	ESP_BOOT0
M2	PC3 *	I/O	GPIO_Output	ESP_RST
M4	VDDA	Power		
M5	VDD	Power		
M6	VSS	Power		
M7	PC5	I/O	FMC_SDCKE0	
M8	PB1	I/O	LTDC_G0	
M9	VDD	Power		
M10	VSS	Power		
M11	PH7	I/O	I2C3_SCL	
M12	PE14	I/O	FMC_D11	
N3	VSSA	Power		
N7	PF11	I/O	FMC_SDNRAS	
N8	PE8	I/O	FMC_D5	
N9	PG1	I/O	FMC_A11	
N10	PF15	I/O	FMC_A9	
N11	PF13	I/O	FMC_A7	
N14	PH10	I/O	LTDC_R4	
P1	PA0	I/O	UART4_TX	
P2	PA1	I/O	UART4_RX	
P4	PH4	I/O	LTDC_G5	

Pin Number UFBGA176	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
P5	PA4	I/O	LTDC_VSYNC	
P8	PG0	I/O	FMC_A10	
P9	PE7	I/O	FMC_D4	
P11	PF12	I/O	FMC_A6	
P12	PE12	I/O	FMC_D9	
P13	PE13	I/O	FMC_D10	
P14	PE15	I/O	FMC_D12	
R1	VSS	Power		
R4	PH3	I/O	LTDC_R1	
R5	PH5	I/O	FMC_SDNWE	
R6	PC4	I/O	FMC_SDNE0	
R8	PB0	I/O	LTDC_G1	
R9	PE10	I/O	FMC_D7	
R10	PF14	I/O	FMC_A8	
R11	PE9	I/O	FMC_D6	
R12	PE11	I/O	FMC_D8	
R13	VCAP	Power		
R14	VDDLDO	Power		
R15	VSS	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H725/735
MCU	STM32H725IGKx
Datasheet	DS13311_Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

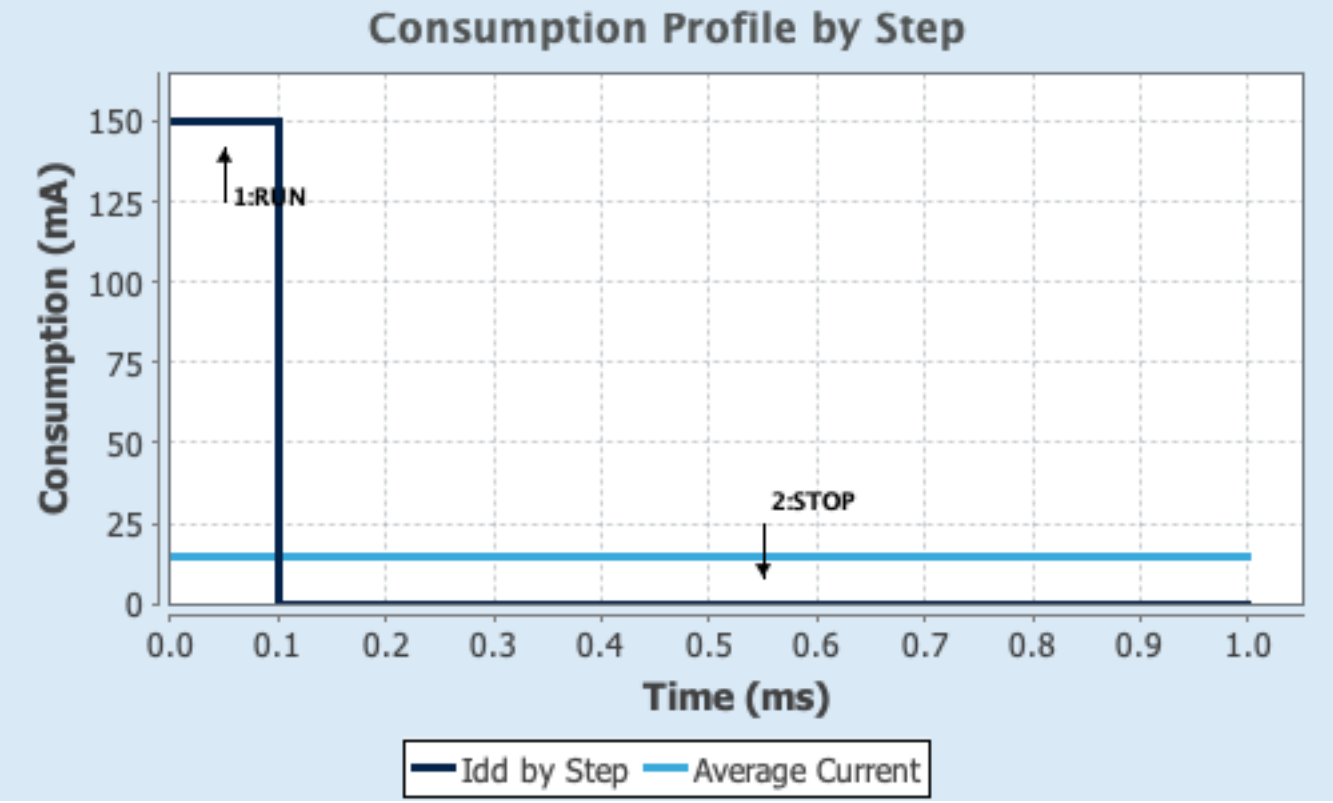
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0/Boost	SVOS3: System-Scale3/SMPS-LDO
D1 Mode	DRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	SRAM1/FlashMode-ON/Cache	NA
CPU Frequency	550 MHz	0 Hz
Clock Configuration	HSE BYP PLL	LSE LowDrive RTC
Clock Source Frequency	8 MHz	32.768 kHz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	150 mA	2.5 µA
Duration	0.1 ms	0.9 ms
DMIPS	1177.0	0.0
Ta Max	107.9	125
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	15 mA
Battery Life	1 day, 17 hours	Average DMIPS	1177.0 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
150
125
100
75
50
25
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	STM32CubeMX
Project Folder	/Users/joshua/ARGUS/Hardware/STM32CubeMX
Toolchain / IDE	EWARM V8.50
Firmware Package Name and Version	STM32Cube FW_H7 V1.12.1
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_FMC_Init	FMC
4	MX_I2C1_Init	I2C1
5	MX_I2C3_Init	I2C3
6	MX_I2C4_Init	I2C4
7	MX_LTDC_Init	LTDC
8	MX_SDMMC2_MMC_Init	SDMMC2
9	MX_SPI4_Init	SPI4
10	MX_TIM3_Init	TIM3
11	MX_UART4_Init	UART4

Rank	Function Name	Peripheral Instance Name
12	MX_UART7_Init	UART7
13	MX_USB_OTG_HS_PCD_Init	USB_OTG_HS

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7

3.1.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode	Enabled

Cortex Interface Settings:

CPU ICache	Disabled
CPU DCache	Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled
MPU Region Base Address	**0x0 ***
MPU Region Size	4GB
MPU SubRegion Disable	**0x87 ***
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	DISABLE
MPU Shareability Permission	ENABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region	Disabled

3.2. DEBUG

Debug: Trace Asynchronous Sw

3.3. FMC

SDRAM 1

Clock and chip enable: SDCKE0+SDNE0

Internal bank number: 4 banks

Address: 12 bits

Data: 16 bits

3.3.1. SDRAM 1:

SDRAM control:

Bank	SDRAM bank 1
Number of column address bits	8 bits
Number of row address bits	12 bits
CAS latency	1 memory clock cycle
Write protection	Disabled
SDRAM common clock	Disabled
SDRAM common burst read	Disabled
SDRAM common read pipe delay	0 HCLK clock cycle

SDRAM timing in memory clock cycles:

Load mode register to active delay	16
Exit self-refresh delay	16

Self-refresh time	16
SDRAM common row cycle delay	16
Write recovery time	16
SDRAM common row precharge delay	16
Row to column delay	16

3.3.2. Bank Mapping:

Mapping parameters:

FMC bank mapping	Default mapping

3.4. I2C1

I2C: I2C

3.4.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x00707CBB ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.5. I2C3

I2C: I2C

3.5.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled

I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x00707CBB ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.6. I2C4

I2C: I2C

3.6.1. Parameter Settings:

Timing configuration:

Custom Timing	Disabled
I2C Speed Mode	Standard Mode
I2C Speed Frequency (KHz)	100
Rise Time (ns)	0
Fall Time (ns)	0
Coefficient of Digital Filter	0
Analog Filter	Enabled
Timing	**0x00707CBB ***

Slave Features:

Clock No Stretch Mode	Disabled
General Call Address Detection	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0

3.7. LTDC

Display Type: RGB888 (24 bits)

3.7.1. Parameter Settings:

Synchronization for Width:

Horizontal Synchronization Width	8
Horizontal Back Porch	7
Active Width	640
Horizontal Front Porch	6
HSync Width	7
Accumulated Horizontal Back Porch Width	14
Accumulated Active Width	654
Total Width	660

Synchronization for Height:

Vertical Synchronization Height	4
Vertical Back Porch	2
Active Height	480
Vertical Front Porch	2
VSync Height	3
Accumulated Vertical Back Porch Height	5
Accumulated Active Height	485
Total Height	487

Signal Polarity:

Horizontal Synchronization Polarity	Active Low
Vertical Synchronization Polarity	Active Low
Data Enable Polarity	Active Low
Pixel Clock Polarity	Normal Input

Layer Default Color:

Red	0
Green	0
Blue	0

3.7.2. Layer Settings:

Layer Default Color:

Layer 0 - Alpha	0
Layer 0 - Blue	0
Layer 0 - Green	0
Layer 0 - Red	0
Layer 1 - Alpha	0
Layer 1 - Blue	0
Layer 1 - Green	0
Layer 1 - Red	0

Number of Layers:

Number of Layers	2 layers

Windows Position:

Layer 0 - Window Horizontal Start	0
Layer 0 - Window Horizontal Stop	0
Layer 0 - Window Vertical Start	0
Layer 0 - Window Vertical Stop	0
Layer 1 - Window Horizontal Start	0
Layer 1 - Window Horizontal Stop	0
Layer 1 - Window Vertical Start	0
Layer 1 - Window Vertical Stop	0

Pixel Parameters:

Layer 0 - Pixel Format	ARGB8888
Layer 1 - Pixel Format	ARGB8888

Blending:

Layer 0 - Alpha constant for blending	0
Layer 0 - Blending Factor1	Alpha constant
Layer 0 - Blending Factor2	Alpha constant
Layer 1 - Alpha constant for blending	0
Layer 1 - Blending Factor1	Alpha constant
Layer 1 - Blending Factor2	Alpha constant

Frame Buffer:

Layer 0 - Color Frame Buffer Start Adress	0
Layer 0 - Color Frame Buffer Line Length (Image Width)	0
Layer 0 - Color Frame Buffer Number of Lines (Image Height)	0
Layer 1 - Color Frame Buffer Start Adress	0
Layer 1 - Color Frame Buffer Line Length (Image Width)	0
Layer 1 - Color Frame Buffer Number of Lines (Image Height)	0

3.8. MEMORYMAP

mode: Activated

3.9. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.9.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_DIRECT_SMPS_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 3

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
CSI Calibration Value	16
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	1 WS (2 CPU cycle)

PLL range Parameters:

PLL1 input frequency range	Between 8 and 16 MHz
PLL3 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range
PLL3 clock Output range	Wide VCO range

3.10. SDMMC2

Mode: MMC 4 bits Wide bus

3.10.1. Parameter Settings:

SDMMC parameters:

Clock transition on which the bit capture is made	Rising transition
SDMMC Clock output enable when the bus is idle	Disable the power save for the clock
SDMMC hardware flow control	The hardware control flow is disabled
SDMMC clock divide factor	0
Is external transceiver present ?	no

3.11. SPI4

Mode: Full-Duplex Master

3.11.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	4 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	**32.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	Enabled
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Tx Crc Initialization Pattern	All Zero Pattern
Rx Crc Initialization Pattern	All Zero Pattern
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled

3.12. SYS

Timebase Source: SysTick

3.13. TIM3

Channel3: PWM Generation CH3

3.13.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	0

Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.14. UART4

Mode: Asynchronous

3.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.15. UART7

Mode: Asynchronous

Hardware Flow Control (RS232): CTS/RTS

3.15.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200

Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	FIFO mode disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.16. USB_OTG_HS
Internal FS Phy: Device_Only
3.16.1. Parameter Settings:

Speed	Device Full Speed 12MBit/s
Enable internal IP DMA	Disabled
Physical interface	Internal Phy
Low power	Disabled
Link Power Management	Disabled
Use dedicated end point 1 interrupt	Disabled
VBUS sensing	Disabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
DEBUG	PB3(JTDO/TRACESWO)	DEBUG_JTDO-SWO	n/a	n/a	n/a	
	PA14(JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
	PA13(JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
FMC	PD1	FMC_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD0	FMC_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG15	FMC_SDNCAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF1	FMC_A1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF0	FMC_A0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG5	FMC_BA1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF2	FMC_A2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG8	FMC_SDCLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG4	FMC_BA0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF4	FMC_A4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF5	FMC_A5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF3	FMC_A3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD14	FMC_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD15	FMC_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD9	FMC_D14	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD10	FMC_D15	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD8	FMC_D13	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC5	FMC_SDCKE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE14	FMC_D11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF11	FMC_SDNRAS	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE8	FMC_D5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG1	FMC_A11	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF15	FMC_A9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF13	FMC_A7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG0	FMC_A10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE7	FMC_D4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF12	FMC_A6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE12	FMC_D9	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE13	FMC_D10	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE15	FMC_D12	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PH5	FMC_SDNWE	Alternate Function Push Pull	No pull-up and no pull-down	Very High	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC4	FMC_SDNE0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE10	FMC_D7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PF14	FMC_A8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE9	FMC_D6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PE11	FMC_D8	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
I2C1	PB6	I2C1_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PB7	I2C1_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
I2C3	PC9	I2C3_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PH7	I2C3_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
I2C4	PD13	I2C4_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PD12	I2C4_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
LTDC	PB8	LTDC_B6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD3	LTDC_G7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA15(JTDI)	LTDC_R3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG13	LTDC_R0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD2	LTDC_B7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC12	LTDC_R6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH14	LTDC_G3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA8	LTDC_B3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB5	LTDC_B5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG14	LTDC_B0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC11	LTDC_B4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC10	LTDC_R2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH13	LTDC_G2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG12	LTDC_B1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD6	LTDC_B2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH15	LTDC_G4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA9	LTDC_R5	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC7	LTDC_G6	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC6	LTDC_HSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG6	LTDC_R7	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PG7	LTDC_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF10	LTDC_DE	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB1	LTDC_G0	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH10	LTDC_R4	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH4	LTDC_G5	Alternate Function Push Pull	No pull-up and no pull-down	Low	

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PA4	LTDC_VSYNC	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PH3	LTDC_R1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB0	LTDC_G1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
RCC	PH0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SDMMC2	PG11	SDMMC2_D2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG9	SDMMC2_D0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PB4(NJTRST)	SDMMC2_D3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PD7	SDMMC2_CMD	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PG10	SDMMC2_D1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PC1	SDMMC2_CK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
SPI4	PE2	SPI4_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE5	SPI4_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PE6	SPI4_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Low	
TIM3	PC8	TIM3_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	LCD_PWM
UART4	PA0	UART4_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA1	UART4_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
UART7	PF6	UART7_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF8	UART7_RTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF7	UART7_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PF9	UART7_CTS	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USB_OTG_HS	PA12	USB_OTG_HS_DP	n/a	n/a	n/a	
	PA11	USB_OTG_HS_DM	n/a	n/a	n/a	
GPIO	PE0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	EEP_WC
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PC13	GPIO_EXTI13	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	ADC_DRDY
	PD4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	SDMMC_CD
	PA10	GPIO_EXTI10	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	TOUCH_IRQ
	PC15-OSC32_OUT	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ADC_SYNC
	PC14-OSC32_IN	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ADC_CS
	PB15	GPIO_EXTI15	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	RTC_INTB
	PB14	GPIO_EXTI14	External Interrupt Mode with	No pull-up and no pull-down	n/a	RTC_INTA

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
			Rising edge trigger detection			
	PC2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ESP_BOOT0
	PC3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	ESP_RST

4.2. DMA configuration

nothing configured in DMA service

4.3. BDMA configuration

nothing configured in DMA service

4.4. MDMA configuration

nothing configured in DMA service

4.5. NVIC configuration

4.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
PVD/AVD through EXTI Line detection Interrupt	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
TIM3 global interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
EXTI line[15:10] interrupts	unused		
FMC global interrupt	unused		
UART4 global interrupt	unused		
I2C3 event interrupt	unused		
I2C3 error interrupt	unused		
USB On The Go HS End Point 1 Out global interrupt	unused		
USB On The Go HS End Point 1 In global interrupt	unused		
USB On The Go HS global interrupt	unused		
FPU global interrupt	unused		
UART7 global interrupt	unused		
SPI4 global interrupt	unused		
LTDC global interrupt	unused		
LTDC Error global Interrupt	unused		
I2C4 event interrupt	unused		
I2C4 error interrupt	unused		
SDMMC2 global interrupt	unused		
HSEM1 global interrupt	unused		

4.5.2. NVIC Code generation

Enabled interrupt Table	Select for init	Generate IRQ	Call HAL handler

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current



6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h72x-3x_line_product-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Security Bulletin	https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4839-level-1-cache-on-stm32f7-series-and-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4990-getting-started-with-sigmadelta-digital-interface-on-applicable-stm32-microcontrollers-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5073-receiving-spdif-audio-stream-with-the-stm32f4f7h7-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5419-getting-started-with-stm32h723733-stm32h725735-and-stm32h730-value-line-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5293-migration-guide-from-stm32f7-series-to-stmh74x75x-stm32h72x73x-and-stmh7a37bx-devices-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5325-how-to-use-the-cordic-to-perform-mathematical-functions-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5927-i3c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5342--how-to-use-

	error-correction-code-ecc-management-for-internal-memories-protection-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5405-how-to-use-fdcan-bootloader-protocol-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4861-introduction-to-lcdtft-display-controller-ltdc-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf

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Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5507-how-to-use-crc-to-check-the-integrity-of-the-internal-flash-memory-on-stm32h7-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5050-getting-started-with-octospi-hexadecaspi-and-xspi-interface-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5337-guidelines-for-estimating-stm32h7-mcus-lifetime-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5348-introduction-to-fdcan-peripherals-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5020-introduction-to-digital-camera-interface-dcmi-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4566-how-to-extend-the-dac-performance-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5647-the-goertzel-algorithm-to-compute-individual-terms-of-the-discrete-fourier-transform-dft-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-introduction-to-

	system-memory-boot-mode-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5200-getting-started-with-the-stm32h7-mcu-sdmmc-host-controller-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4323-getting-started-with-stemwin-library-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4435-guidelines-for-obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4891-stm32h72x-stm32h73x-and-singlecore-stm32h74x75x-system-architecture-and-performance-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5001-stm32cube-expansion-package-for-stm32h7-series-mdma-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5014-stm32h7x3-smart-power-management-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5361-getting-started-

for related Tools & Software	with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
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